2026 QS WORLD UNIVERSITY RANKINGS DATA ANALYSIS

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INTRODUCTION TO DATASET



DATA OVERVIEW

- **Source:** QS World University Rankings 2026 dataset (1,501 institutions, 30 variables)
- **Timeframe:** Rankings for year 2026 (with prior year comparison)
- **Coverage:** Institutions from multiple countries and regions worldwide
- **Structure:**
  - ***Categorical:*** *Institution name, country, region, size, focus, status, ranks*
  - ***Numerical:*** *Academic reputation, employer reputation, faculty/student ratio, international indicators, sustainability, overall score*
- **Data Shape:** 1,501 rows , 30 columns
- **Missing Data:** Present in several ranking and score columns, mostly < 8% missing per column



BACKGROUND INFORMATION AND PURPOSE OF THE ANALYSIS



BACKGROUND INFORMATION

- **Source:** The data was sourced directly from the **official QS World University Rankings 2026.**
- The universities dataset *includes published ranking data* for over 1,500 institutions worldwide (evaluated by QS Quacquarelli Symonds).
- **Foundation of the Data:** The QS rankings *are based on global surveys and institutional data*, and the scoring methodology includes employer reputation, academic reputation, research citations, internationalization, and sustainability.
- **Rankings Include:** Current year rank, previous year rank, and ranks for each sub-score metric
- **Scores Include:** Numeric indicators for each ranking metric (e.g., AR SCORE for Academic Reputation, IRN SCORE for International Research Network)
- **Real-World Use:**
  - *Utilized by prospective students for decision-making*
  - *Utilized by institutions for benchmarking performance*
  - *Utilized by policymakers and researchers to assess global higher education trends*



PURPOSE OF ANALYSIS

- **Objective:** Explore, clean, and analyze ranking data to identify trends, patterns, and outliers in global university performance
- **Key Questions:**
  - *Which factors are most strongly associated with overall performance?*
  - *How do top-ranked universities differ from lower-ranked ones?*
  - *Are there notable outliers in key metrics?*
- **Planned Outcomes:**
  - *Identify improvement areas for educational institutions*
  - *Provide actionable recommendations for strategic development*
  - *Present insights visually for decision-making and stakeholder reporting*



DATA CLEANING PROCESS



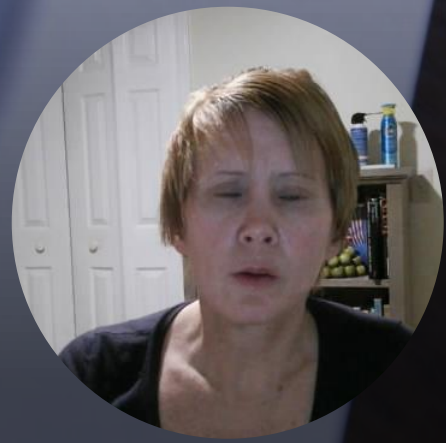
DATA CLEANING PROCESSES

Data Cleaning Steps

- **Handling Missing Values**
  - Some missing values were identified across the dataset.
  - *Numeric fields (e.g., AR SCORE, ER SCORE, FSR SCORE, etc.) were imputed using the median* because it is less sensitive to extreme values than the mean.
  - This ensured that the replacement values *reflected the general distribution of scores* without being skewed by top-performing universities like MIT.
  - *Categorical fields (e.g., Region, Focus, Status) were imputed using the mode*, since it represents the most frequent category and helps preserve the most common institutional characteristics.
- **Removing Duplicates**
  - Verified that *each university appeared only once* in the dataset.
  - No duplicates were detected, *ensuring all records were unique*.
- **Handling Outliers**
  - Outliers *were identified in score distributions* (institutions scoring close to 100 versus those scoring close to 1).
  - These were determined to be *true reflections* of institutional performance rather than data errors.
  - *Outliers were retained* because removing them would distort the reality of global university rankings by minimizing ge performance gaps between top-ranked and lower-ranked educational institutions.



KEY STATISTICS ANALYSIS



PERFORMANCE SCORES IN THE ANALYSIS

- The **QS** dataset consists of several performance scores, each measuring a different dimension of university quality:
- **AR**: Academic Reputation
- **ER**: Employer Reputation
- **FSR**: Faculty-Student Ratio
- **CPF**: Citations per Faculty
- **IFR**: International Faculty Ratio
- **IRN**: International Research Network
- **EO**: Employment Outcomes
- **SUS**: Sustainability
- **ISD**: International Student Diversity
- **Overall Score**: a composite measure combining all the scores (with different weights).



SUMMARY OF KEY STATISTICS

- **Key Insights**
- **Internationalization is the biggest differentiator**
  - Huge variance in IFR, ISR, ISD *shows universities either excel or barely engage* in global faculty/student diversity.
  - Strong IRN scores suggest that *collaborative research is easier to achieve than true internationalization of campus life*.
- **Sustainability is becoming a baseline**
  - With a mean of 51, *it's one of the highest-scoring metrics*, meaning institutions are actively investing in ESG (Environmental, Social, and Governance) initiatives.
- **Prestige and employability are concentrated in a small group of universities**
  - AR and ER have low averages and high variance.
  - This means the "halo effect" benefits only a small set of elite
- **Student outcomes are weaker than research outputs**
  - EO mean is low (30), *suggesting many universities underperform* to employability.



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The dataset includes descriptive info (country, size, status) and performance scores.
For this analysis, we narrow down to the **11 main score columns** that drive the ranki

[22]:

```
# Select relevant score columns

# We extract only the performance score columns to focus
# the analysis on quantitative metrics (excluding metadata).
score_cols = [
    "AR SCORE", "ER SCORE", "FSR SCORE", "CPF SCORE",
    "IFR SCORE", "ISR SCORE", "ISD SCORE", "IRN SCORE",
    "EO SCORE", "SUS SCORE", "Overall SCORE"
]
```

[23]:

```
# Summary statistics for all institutions

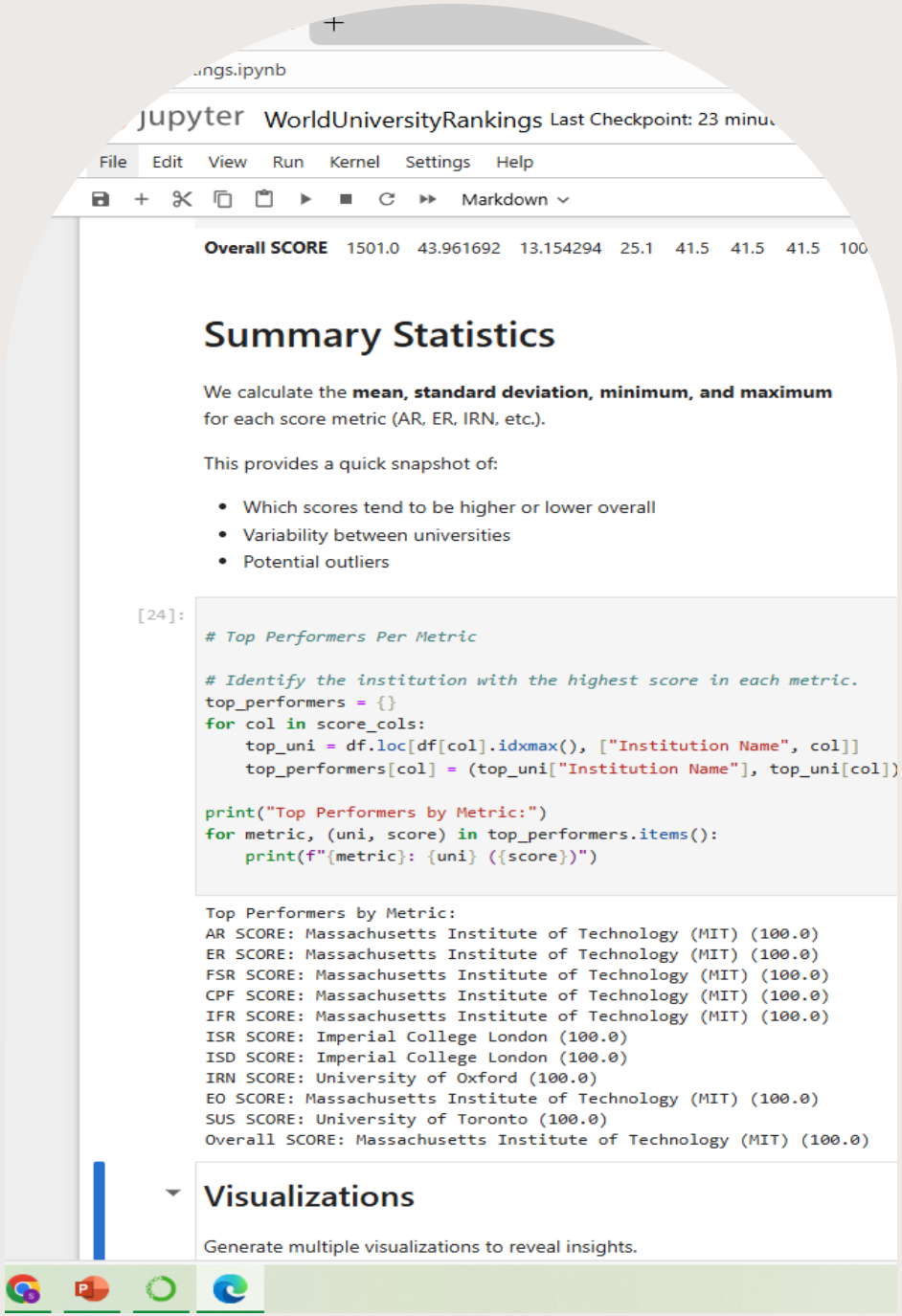
# Generate descriptive statistics (mean, min, max, std) for
# each score metric to understand overall distribution.
summary_stats = df[score_cols].describe().T
display(summary_stats)
```

	count	mean	std	min	25%	50%	75%	max
AR SCORE	1501.0	25.785943	24.500905	1.0	8.8	16.0	32.7	100.0
ER SCORE	1501.0	26.944237	25.504494	1.0	8.5	16.5	37.5	100.0
FSR SCORE	1501.0	33.950433	28.440071	1.0	10.8	23.5	50.5	100.0
CPF SCORE	1501.0	30.425516	29.679882	1.0	6.0	18.0	49.7	100.0
IFR SCORE	1501.0	35.366356	34.423503	1.0	7.0	20.1	61.9	100.0
ISR SCORE	1501.0	32.982145	32.416254	1.0	6.0	19.4	55.6	100.0
ISD SCORE	1501.0	34.210793	30.786886	1.0	8.8	21.7	54.5	100.0
IRN SCORE	1501.0	53.360160	28.901483	1.0	27.6	55.8	78.4	100.0
EO SCORE	1501.0	29.989674	29.197573	1.0	6.2	17.9	46.0	100.0
SUS SCORE	1501.0	51.213524	21.097949	3.0	35.9	48.7	66.1	100.0
Overall SCORE	1501.0	43.961692	13.154294	25.1	41.5	41.5	41.5	100.0

ACTIONABLE RECOMMENDATIONS (BASED ON KEY STATISTICS)

- **For mid-tier universities:**
  - *Focus on niche internationalization* by building partnerships in *underrepresented regions and attracting diverse faculty* through joint appointments.
  - *Improve employer engagement* with internship programs, mentorship, and corporate partnerships.
- **For research universities:**
  - *Leverage strong* IRN and CPF to build global research hubs.
  - *Invest in translating research prestige* into graduate employability branding.
- **For policy makers:**
  - *Encourage incentives for sustainability* to continue its positive trend.
  - *Support universities* in global mobility programs (faculty exchanges, scholarships).





TOP 10 PERFORMING UNIVERSITIES

- **Interpretation of Top Performers**
  - ***Results:***
    - **MIT** dominates most categories (AR, ER, FSR, CPF, IFR, EO, Overall).
  - ***Imperial College London*** *leads in international student/faculty ratios (ISR, ISD).*
  - ***University of Oxford*** *tops international research collaboration (IRN).*
  - ***University of Toronto*** *excels in sustainability (SUS).*

- **Key Insights**
  - ***MIT is the global benchmark***
    - Excels not just in reputation but across structural aspects (faculty ratios, research, and employability).
    - Its holistic dominance shows a synergy between academic prestige, research power, and career outcomes.
  - ***Imperial College London leads in global diversity***
    - Its appeal to international students/faculty highlights the strength o
      global education hub.
  - ***Oxford reinforces research leadership***
    - Its strong IRN score aligns with its long history of global aca
  - ***Toronto sets sustainability standard***
    - Shows how Canadian institutions leverage strong ESG polici
      engagement



ACTIONABLE RECOMMENDATIONS (BASED ON TOP PERFORMING UNIVERSITIES)

- **Universities aspiring to global leadership:**
  - Benchmark against *MIT's integrated model* — invest simultaneously in faculty excellence, research funding, and employer partnerships.
- **Universities in Europe/Asia/Africa:**
  - Follow *London Imperial's strategy* of building a strong international brand through faculty/student recruitment pipelines.
- **Universities with strong research but weak global ties:**
  - Emulate *Oxford* by prioritizing strategic alliances and research consortia.
- **Universities with regional leadership ambitions:**
  - Position themselves like *Toronto*, using sustainability and community impact as brand-defining st



DATA VISUALIZATIONS



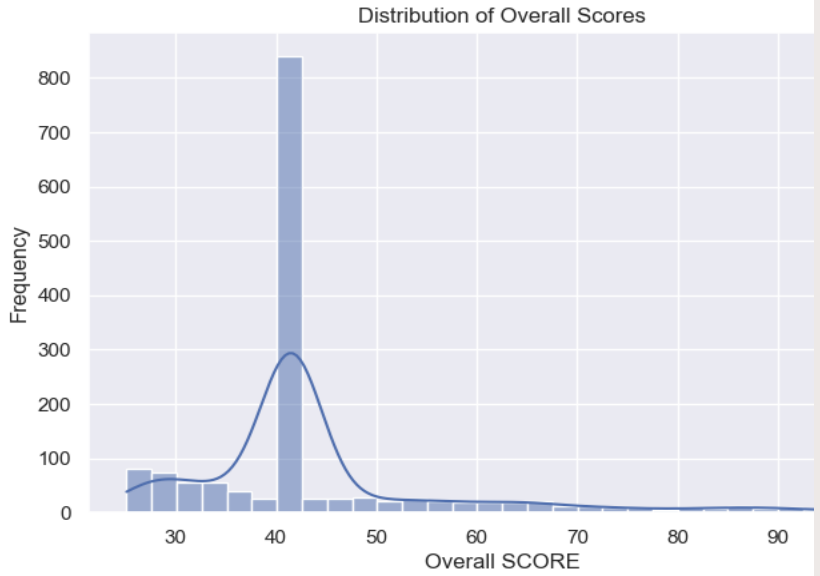
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- Distribution of scores
- Top universities
- Regional differences
- Correlations between metrics

```
# Distribution of Overall Scores
plt.figure(figsize=(8,5))
sns.histplot(df['Overall SCORE'], kde=True, bins=30)
plt.title('Distribution of Overall Scores')
plt.xlabel('Overall SCORE')
plt.ylabel('Frequency')
plt.show()
```

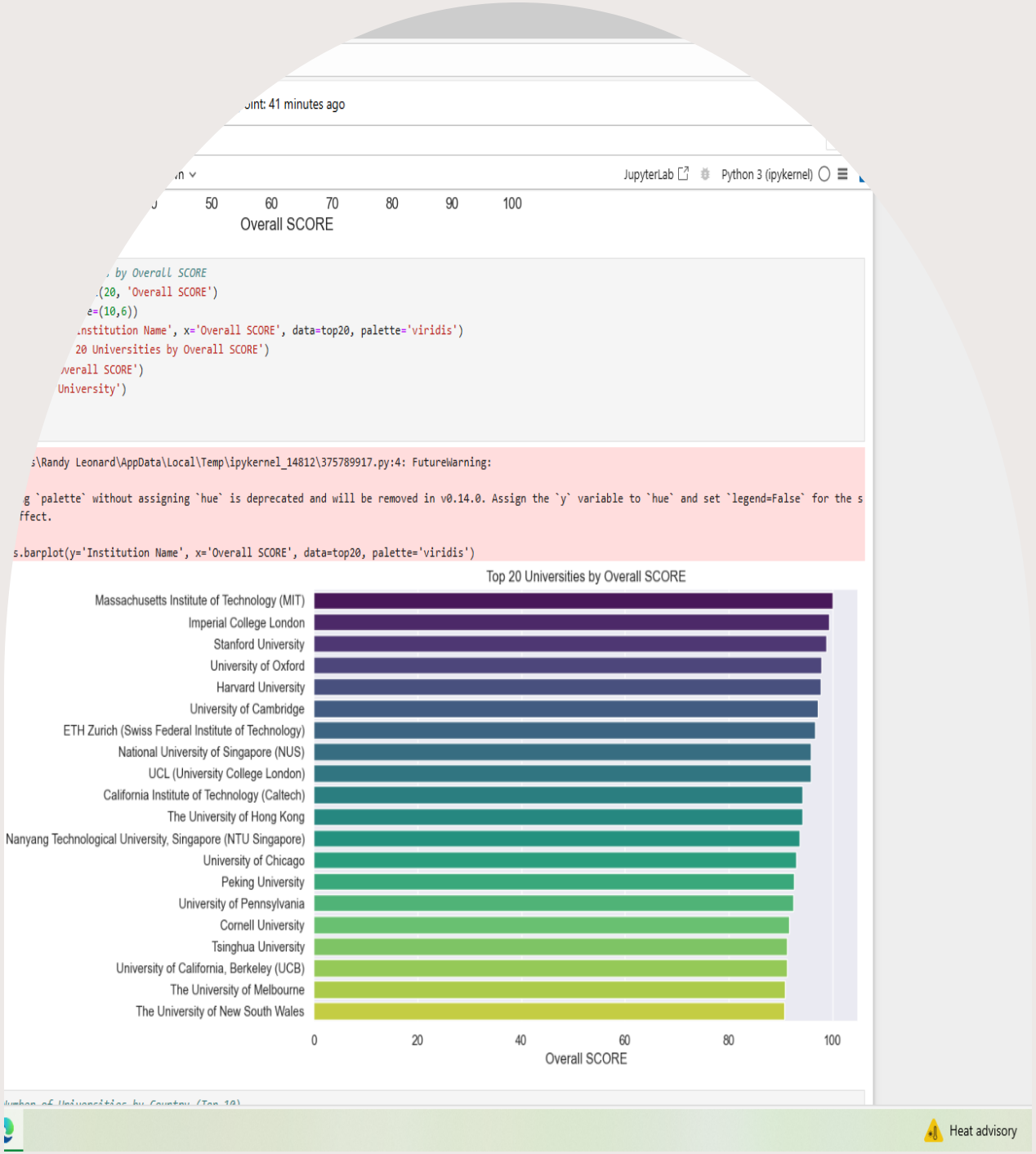



```
# Top 20 Universities by Overall SCORE
top20 = df.nlargest(20, 'Overall SCORE')
plt.figure(figsize=(10,6))
sns.barplot(y='Institution Name', x='Overall SCORE', data=top20, palette='viridis')
plt.title('Top 20 Universities by Overall SCORE')
plt.xlabel('Overall SCORE')
```

DISTRIBUTION OF OVERALL SCORES

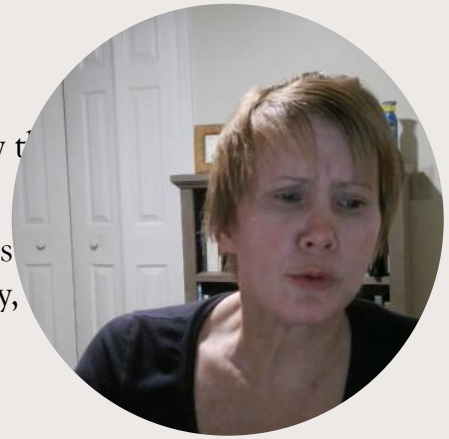
- **Interpretation:**
- The histogram shows that most universities have an overall score tightly clustered around **40**.
- **Actionable Insights:**
- The distribution is highly **right-skewed** — many universities are in the mid-to-low range, while only a few excel significantly.
- **Actionable Recommendations:**
- Consider normalizing or transforming the data if performing statistical modeling to reduce skewness.
- Segment universities into performance tiers (e.g., Elite, High-per… tier, Low-tier) for more meaningful benchmarking.

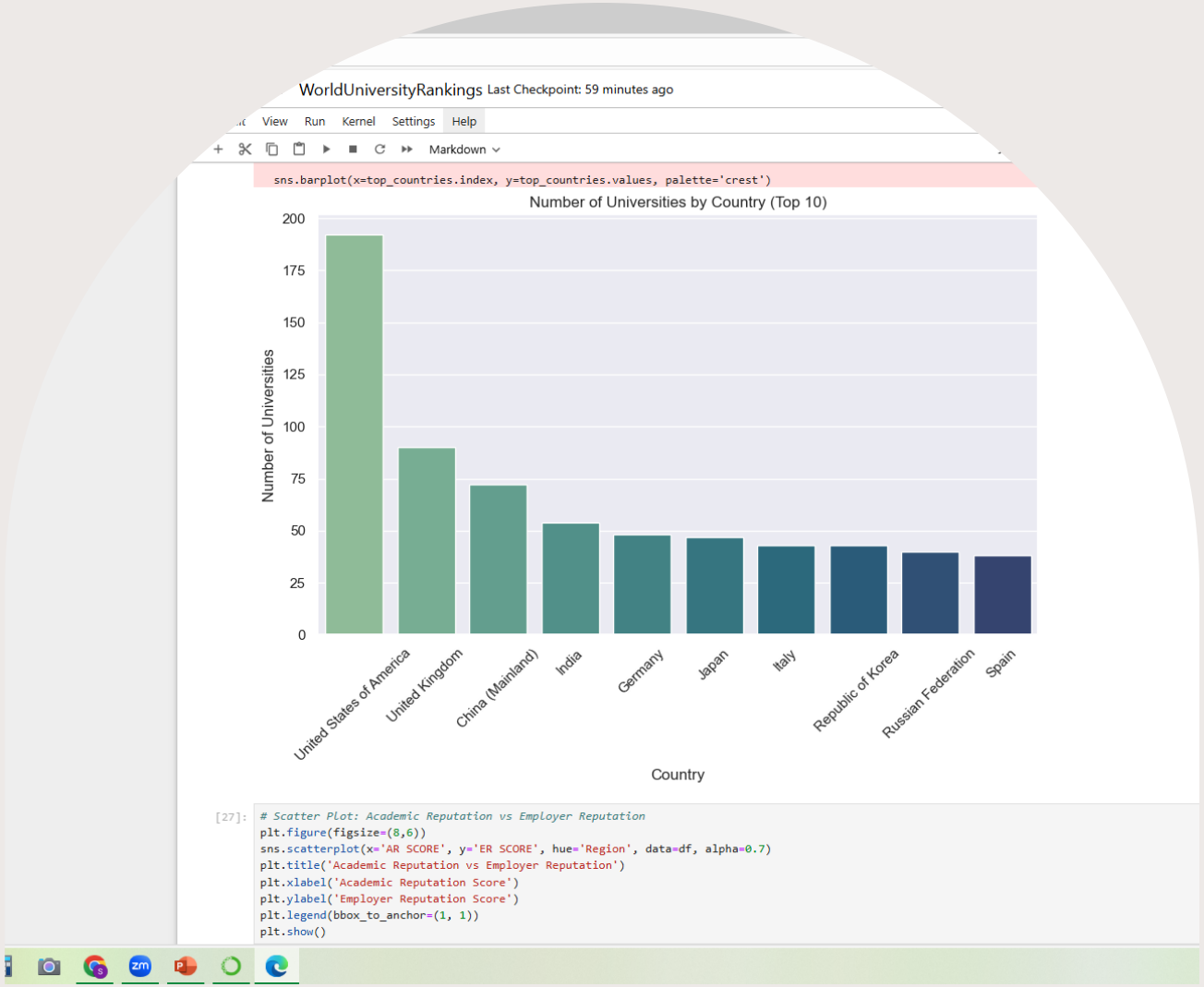




Top 20 Universities by Overall Score

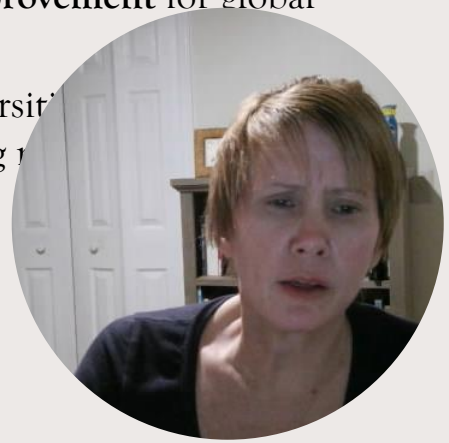
- **Interpretation:**
  - **MIT, Imperial College London, Stanford, Oxford, Harvard, and Cambridge** dominate the top ranks.
  - Many top performers are in the **US and UK**, but there's notable representation from Singapore, Switzerland, China, and Australia.
- **Actionable Insights:**
  - This confirms the **global concentration of elite education in a few countries**.
  - Asian universities like NUS, NTU, and Tsinghua are increasingly competitive.
- **Actionable Recommendations:**
  - For institutions outside the top tier, study t
    output of these leading universities.
  - Governments and education policymakers
    countries might invest in academic quality,
    faculty recruitment to improve rankings.

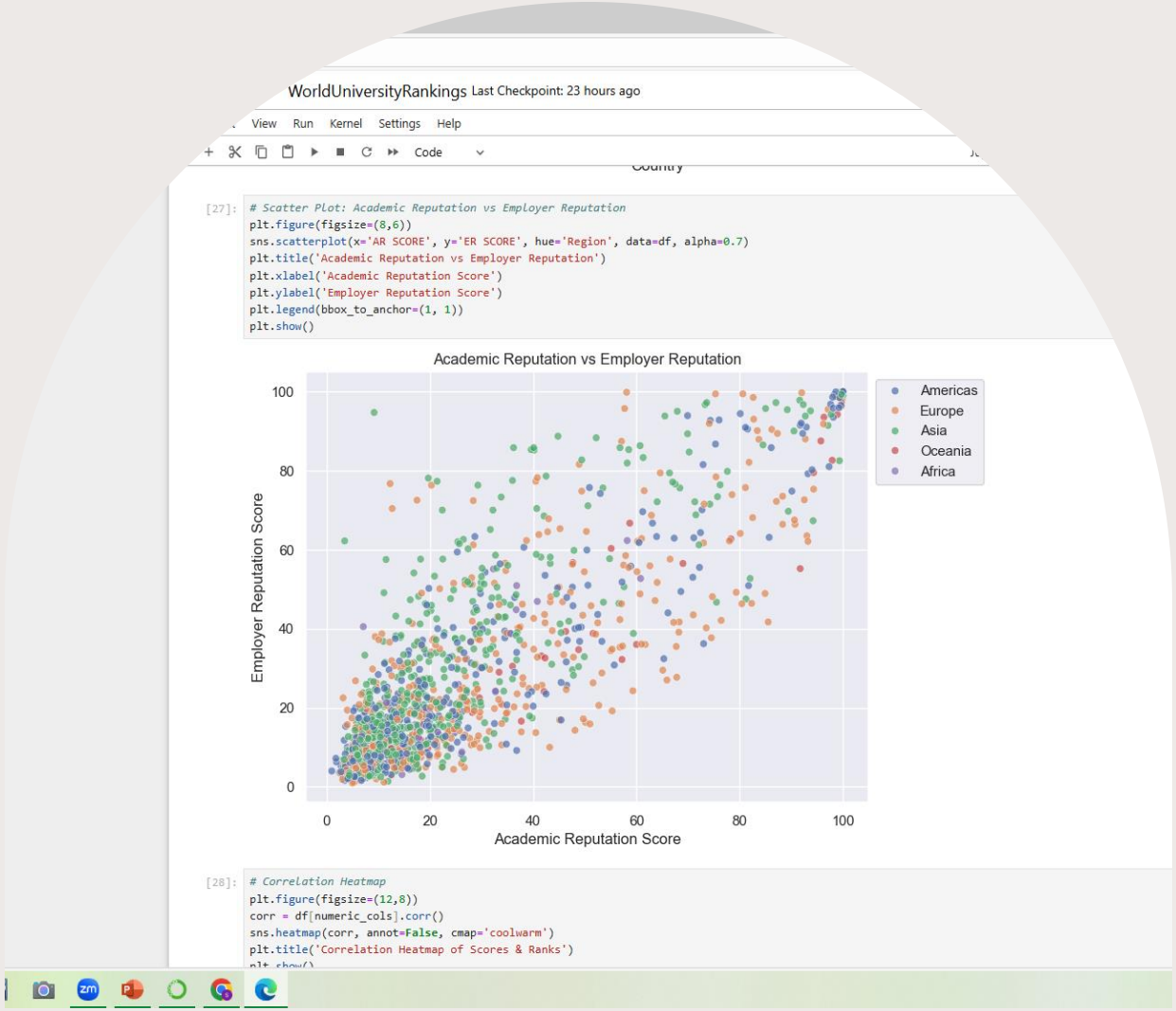




NUMBER OF UNIVERSITIES BY COUNTRY (TOP 10)

- **Interpretation:**
  - The **United States** has the most universities in the ranking (almost 200), followed by the **UK** and **China**.
  - Other countries like India, Germany, and Japan also have sizable representation.
- **Insights:**
  - This chart shows **quantity**, not necessarily quality. The US dominates in representation, which also explains why many of the top 10 universities are from there.
  - China's high count is notable, showing rapid growth in global academic presence.
- **Recommendations:**
  - Countries with many universities (like the US, China, India) should focus on **quality improvement** for global competitiveness.
  - Countries with fewer ranked universit
    increase representation by meeting

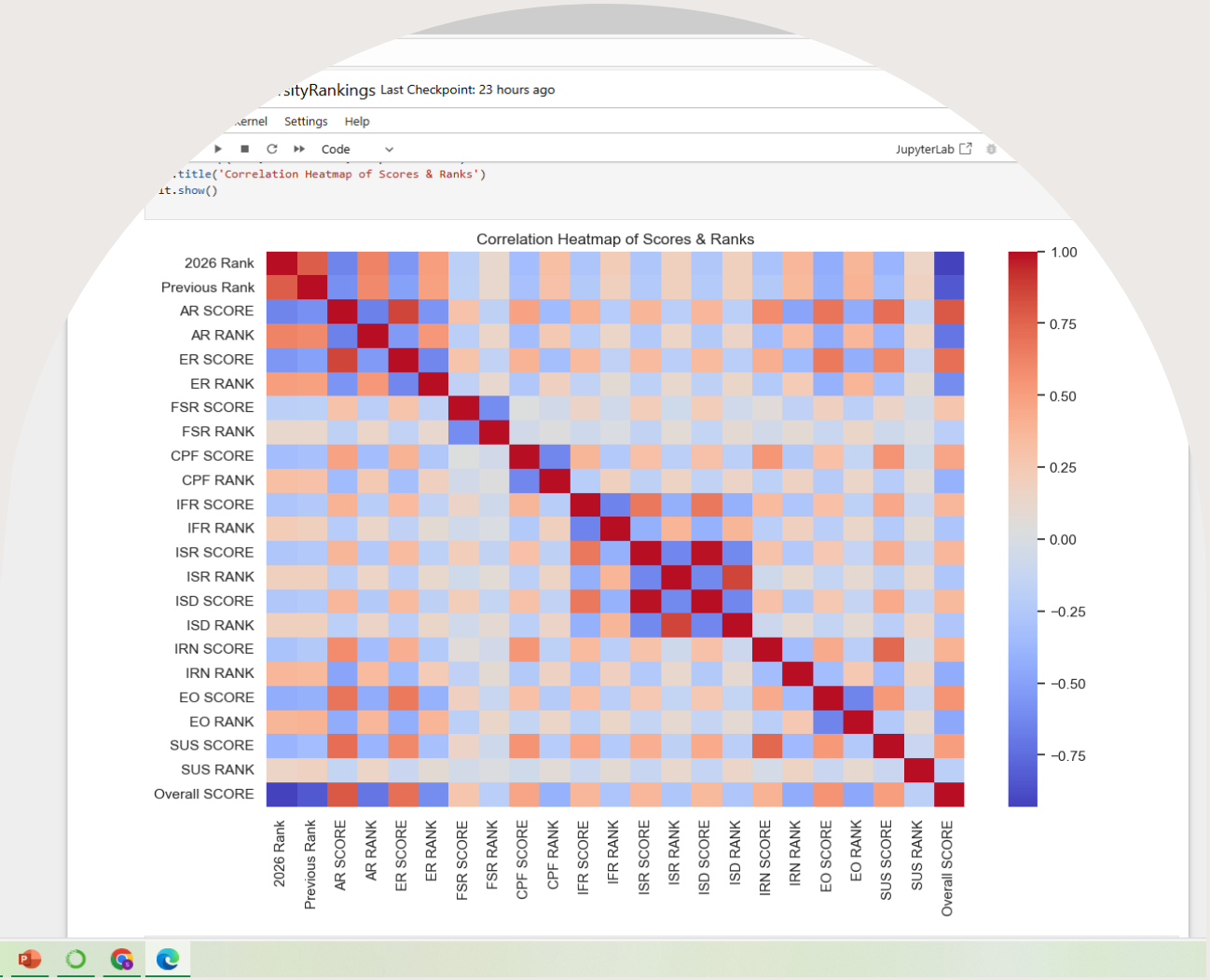




ACADEMIC REPUTATION VS EMPLOYER REPUTATION

- **Interpretation**
  - Positive correlation: universities with high **academic reputation (AR)** also tend to score high on **employer reputation (ER)**.
  - European & American institutions cluster at the top.
  - African universities mainly fall into the lower AR/ER zone.
- **Actionable Insights**
  - Academic prestige **directly influences employability outcomes**.
  - Institutions with weak AR scores **struggle to gain employer trust**.
  - Regional disparities suggest **structural inequality in visibility and resources**.
- **Actionable Recommendations**
  - Invest in **faculty quality & research visibi** (international journals, conferences).
  - Build **industry partnerships** and ex to boost ER.
  - Launch **global marketing campaig** institutional reputation.

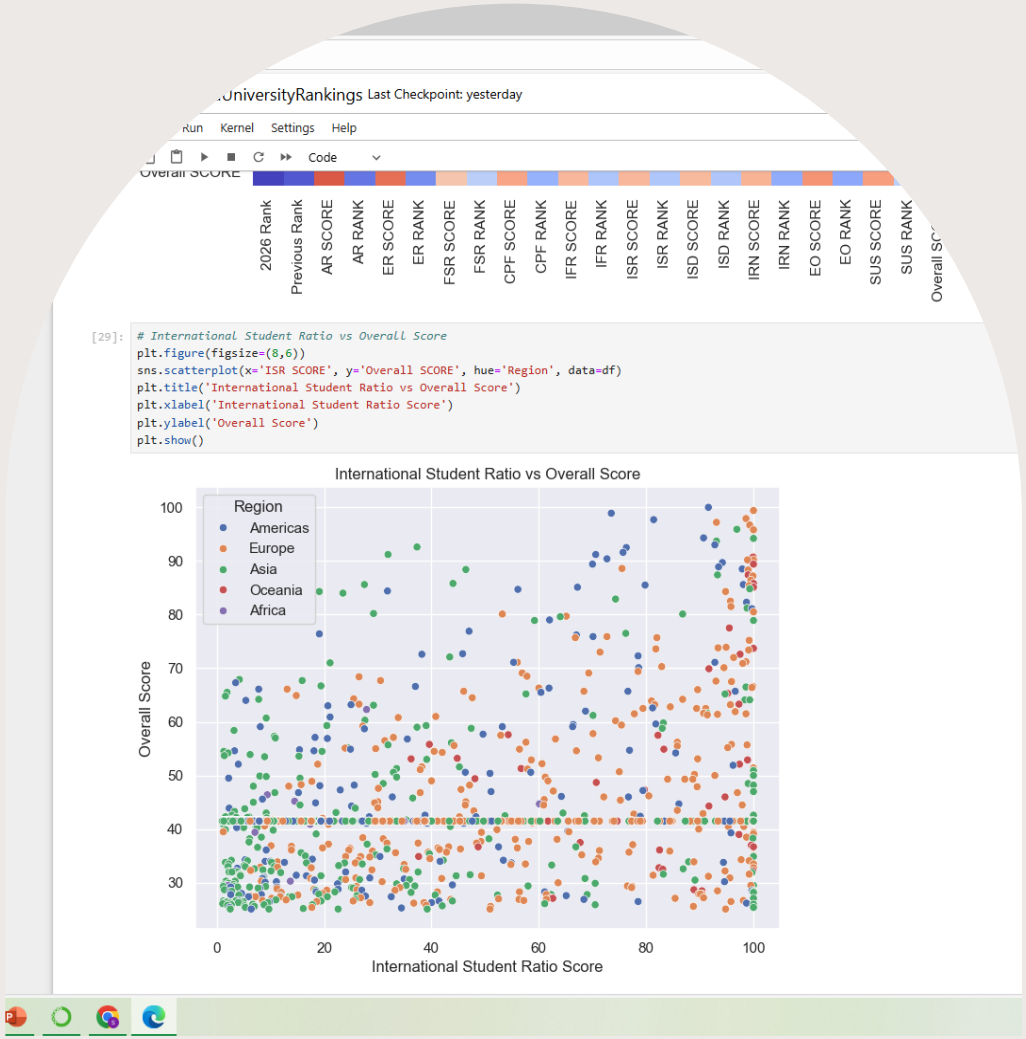




CORRELATION HEATMAP OF SCORES & RANKS

- **Interpretation**
  - **AR and ER scores** show the strongest correlation with the Overall Score.
  - FSR (faculty/student ratio), IRN (international research network), and ISR (international student ratio) have moderate influence.
- **Actionable Insights**
  - Reputation-based measures **weigh the most in ranking outcomes**.
  - Structural/operational factors (e.g., ISR, IRN) are supportive but secondary.
  - Focusing only on operational improvements will not significantly change rankings.
- **Actionable Recommendations**
  - Prioritize **brand-building and visibility strategies** to improve AR and ER.
  - Simultaneously **invest in teaching quali** improve student experience.
  - Track **weaker contributing metrics** a **portfolio approach**.





INTERNATIONAL STUDENT RATIO VS OVERALL SCORE

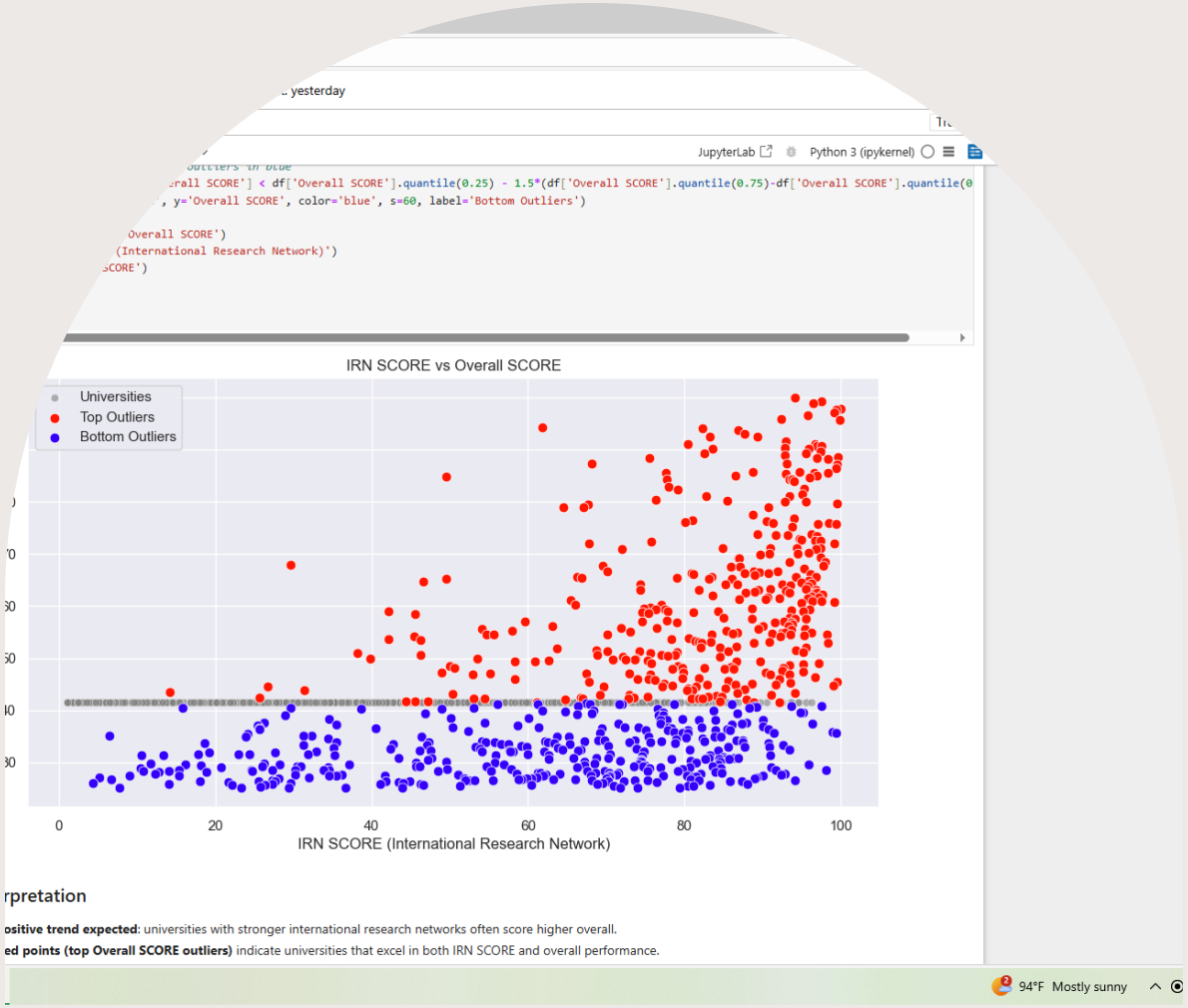
- **Interpretation:**
  - **No strong linear relationship (ISR is not a strong predictor of performance)**
    - Universities with **high ISR (close to 100)** appear across the full range of Overall Scores (30–100).
    - Likewise, schools with **low ISR (close to 0)** can still have high or low Overall Scores.
  - ***Clusters at "40 Overall Score" (reinforce IRS is not a predictor of Overall Score)***
    - The **straight grey line** around Overall Score = **40** indicates that many schools are assigned the same normalized score (a ranking band).
- **Regional differences**
  - **European & Oceania universities (orange, red)** dominate the **high ISR & high Overall Score** quadrant.
  - **Asian universities (green)** have a widespread, often high Overall Score but lower ISR.
  - **Americas (blue)** show moderate ISR but can still achieve strong Overall Scores.
- **Actionable Insights:**
  - **Diversity vs. Ranking**
    - Having a high proportion of international students **does not guarantee** a higher Overall Score.
    - However, top universities (80+ Overall Score) often balance **strong academic reputation + moderate-to-high ISR**.
  - ***Regional Strategy***
    - Universities in **Asia and the Americas** can achieve high rankin… **high ISR** , indicating they rely more on Academic Reputatio…, or Employer Reputation.
    - **European and Oceania institutions** leverage ISR as a com…
  - ***Benchmarking Opportunity***
    - Institutions with **low ISR but high Overall Scores** (especia… **models for prioritizing other KPIs**.
    - However, those with **high ISR but low Overall Scores** may nee… **academic/research factors**.



ACTIONABLE RECOMMENDATIONS BASED ON STUDENT RATIO VS OVERALL SCORE

- **For Universities Seeking to Climb Rankings**
  - *Focus first on academic reputation, Faculty/Student Ratio, and research output*
  - *Use ISR strategically as secondary differentiator.*
- **For Policymakers/Education Planners:**
  - *Balance ISR initiatives with academic investments.*
- **For Marketing Offices and Recruiters:**
  - *Universities in regions with low ISR should focus on academic excellence and growing global outreach rather than ISR alone.*
  - *Highlight ISR only if other ranking drivers are strong.*





INTERNATIONAL RESEARCH NETWORK (IRN) VS OVERALL SCORE

- **Interpretation**
  - *Institutions with higher **IRN scores** generally perform better overall.*
  - *Outliers exist: strong IRN does not always translate to high overall performance.*
  - *Regional divide: **Europe & Americas dominate high IRN/high score zones**, while **Africa is concentrated at lower ranges**.*
- **Actionable Insights**
  - *Strong research networks **boost institutional reputation & collaboration opportunities**.*
  - *IRN alone isn't enough – reputation and teaching quality remain critical.*
  - *Underperforming regions need more access to international collaborations.*
- **Actionable Recommendations**
  - *Build **cross-border research collabo** top-ranked global universities.*
  - *Encourage **joint publications, ex** **programs, and international c***
  - *For developing regions: **form cons** **global institutions** to raise visibili*



OVERALL INSIGHTS SUMMARY

- **Cluster Patterns:**
  - Several universities cluster *around score thresholds (overall score = 40),* indicating bundled rankings.
  - Internationalization (ISR, IRN) *is a not a strong predictor (alone).*
- **Varying Ranking Drives:**
  - Faculty/Student ratio, academic reputation, and employer reputation *consistently correlate with higher overall scores.*
- **Regional Trends:**
- Asia: Higher performance *even though Asian universities have a lower ISR*
- America: Mixed performance, reputation-driven success
- Europe and Oceania: Strong ISR presence, *frequently pulled with higher scores*



SUMMARY OF ACTIONABLE RECOMMENDATIONS

Strengthen Core Academic Drivers

- **Faculty–Student Ratio (FSR):**
  - Hire additional faculty and reduce class sizes
  - Invest in teaching quality through training & incentives
- **Research Output (Research, AR):**
  - Fund interdisciplinary, high-impact research
  - Encourage global collaborations & industry partnerships

Balanced Globalization Strategy

- Utilize international student recruitment (ISR, IRN) as a complement, not a substitute, for quality.

Region-Specific Approaches

- **Americas: Asia & Americas: Climbing Through Academic Excellence**
- Focus on **core quality metrics** (AR, ER, FSR) to rival Europe's historic strength.
  - **Asia:**
  - Invest in top-tier research facilities & global partnersh
  - Enhance faculty recruitment to improve student ratio
  - **Americas:**
  - Address affordability & access to maintain global tale
  - Leverage strong alumni/employer networks to boost em (ER)



SUMMARY OF ACTIONABLE RECOMMENDATIONS

Region-Specific Approaches:

Europe & Oceania: Leveraging International Diversity

- **Europe:**
- Continue Erasmus+ mobility & cross-border research projects
- Maintain leadership in international student/faculty ratios
- **Oceania (Australia/NZ):**
- Expand international partnerships, especially in Asia-Pacific
- Safeguard mobility policies post-pandemic recovery

Policy Incentives for Sustainability

- **How Policymakers Can Boost SUS Scores:**
  - **Funding:** Green campus bonds, sustainability-linked re
  - **Standards:** Require ESG reporting, align accreditatio (Sustainable Development Goals)
  - **Recognition:** National "Green University Awards"



SUMMARY OF ACTIONABLE RECOMMENDATIONS

Policy Incentives for Global Mobility

- **How Policymakers Can Support IFR, ISR, IRN**
  - **Visas:** Simplify & fast-track for students, faculty, researchers
  - **Scholarships:** Fund inbound & outbound mobility (ex: study abroad)
  - **Research Collaboration:** Seed funding for joint projects & centers
  - **Integration:** Support services for language, mentorship, internships

Combined Leverage

- **Linking Sustainability + Mobility**
  - Green mobility scholarships for SDG (Sustainability-r… studies
  - International research hubs focused on climate
  - ESG-focused global partnerships



NEXT STEPS OF ACTION

Deeper Statistical Modeling

- Run regression/correlation analyses across all score metrics to quantify influence on Overall Score.

Institutional Benchmarking

- Identify top-performing universities in each region and analyze their strategic mix of ISR, faculty, and research metrics.
- Display each of the Top 10 universities strongest and weakness areas (scores).

Strategic Prioritization

- Create region-specific playbooks for universities aiming to improve their global ranking.

Data Storytelling Enhancement

- Refine visualizations with trendlines, regression fits, and annotations to clarify actionable findings.



CONCLUSION

- **No Single Silver Bullet**: No single metric (like ISR) *guarantees higher scores* — success *requires a multi-dimensional* strategy.
- **Academic Quality First**: Strong academic reputation and faculty/student balance remain the **foundation of rankings**.
- **Internationalization as Differentiator**:
  - Useful but secondary
  - ISR *helps amplify ranking* only when *combined* with academic strength.
- **Strategic Opportunity**: Universities in Asia and the Americas *can climb further* by **adopting selective internationalization** *without losing focus* on academic excellence.

- **Pathway to Stronger Rankings**
  - **Universities:** Strengthen academic excellence & diversify talent
  - **Regions:** Leverage strengths (Asia/Americas = excellen
    Europe/Oceania = diversity)
  - **Policymakers:** Incentivize sustainability and mobil



THANK YOU!



